Points to Remember

- Crop improvement is the development of improved crop varieties possesing higher yield, better quality, resistance to diseases and shorter duration.
- When breeding takes place between animals of the same breed, it is called inbreeding.The cross between different breeds is called outbreeding.
- The superiority of the hybrid obtained by cross breeding is called as heterosis or hybrid vigour.
- Genetic engineering is the manipulation and transfer of genes from one organism to another organism.
- Stem cells are undifferentiated or unspecialised mass of cells and can be used for the treatment known as stem cell therapy.



TEXTBOOK EVALUATION

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I Choose the correct answer

1. Which method of crop improvement can be practised by a farmer if he is inexperienced?
   a. clonal selection b. mass selection
   c. pureline selection d. hybridisation
2. Pusa Komal is a disease resistant variety of _____.
   a. sugarcane b. rice
   c. cow pea d. maize
3. Himgiri developed by hybridisation and selection for disease resistance against rust pathogens is a variety of _____.
   a. chilli b. maize
   c. sugarcane d. wheat
4. The miracle rice which saved millions of lives and celebrated its 50th birthday is _____
   a. IR 8 b. IR 24 c. Atomita 2 d. Ponni
5. Which of the following is used to produce products useful to humans by biotechnology techniques?
   a. enzyme from organism
   b. live organism
   c. vitamins
   d. both (a) and (b)
6. We can cut the DNA with the help of
   a. scissors b. restriction endonucleases
   c. knife d. RNAase
7. rDNA is a
   a. vector DNA
   b. circular DNA
   c. recombinant of vector DNA and desired DNA
   d. satellite DNA
8. DNA fingerprinting is based on the principle of identifying ---------- sequences of DNA
   a. single stranded b. mutated
   c. polymorphic d. repetitive
9. Organisms with modified endogenous gene or a foregin gene are also known as
   (a) transgenic organsims
   (b) genetically modified
   (c) mutated
   (d) both a and b

10th_Science Unit-20.indd 296 12/3/2022 3:10:40 PM